

Comprehensive Comparative Analysis of Variational Quantum Classifier and Quantum Kernel SVM under NISQ Noise with Classical Baseline and Statistical Significance Evaluation

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Abstract—Quantum Machine Learning (QML) offers the potential to solve data-driven tasks using quantum computing, especially in the NISQ era. However, the impact of noise on quantum models and their comparison with classical baselines remain underexplored. This study performs a statistically rigorous evaluation of two prominent QML models—Variational Quantum Classifier (VQC) and Quantum Kernel Support Vector Machine (QSVM)—using Qiskit Aer under depolarizing and amplitude damping noise models. Ten independent trials were conducted at four noise levels (0, 0.01, 0.05, 0.1), and results were benchmarked against a classical SVM (RBF kernel). QSVM achieved 70% accuracy in ideal settings, while VQC showed better resilience under moderate noise ($59\% \pm 0.11$ at 0.05 noise). Classical SVM consistently achieved 100% accuracy. These findings highlight that while QSVM performs better in low-noise regimes, VQC is more noise-adaptive. This analysis offers essential insights for designing robust QML models for future NISQ-era quantum hardware.

Keywords—quantum machine learning (QML), variational quantum classifier (VQC), quantum kernel support vector machine (QSVM), noisy intermediate-scale quantum (NISQ), qiskit aer, noise-resilience, statistical significance, classical SVM baseline

I. INTRODUCTION

Quantum Machine Learning (QML) has emerged as a promising approach that leverages quantum computation to enhance data-driven tasks, particularly in classification, optimization, and pattern recognition domains [1], [2]. With the advent of Noisy Intermediate-Scale Quantum (NISQ) hardware, hybrid quantum-classical models such as the Variational Quantum Classifier (VQC) and Quantum Kernel Support Vector Machine (QSVM) have gained substantial traction [3], [4]. These models aim to exploit the power of quantum states and entanglement for learning tasks. The VQC relies on Parameterized Quantum Circuits (PQCs) optimized using classical techniques, while the QSVM utilizes quantum-enhanced feature maps to compute similarity kernels in high-dimensional Hilbert spaces [5].

Despite advancements, deploying these models in real-world NISQ environments faces several challenges. Many existing studies evaluate VQC and QSVM under ideal, noise-free simulations without accounting for realistic hardware-

induced noise such as depolarizing and amplitude damping effects [6], [15]. Furthermore, these studies often neglect statistical robustness, providing only single-run accuracy results without confidence intervals or significance testing [16], [17]. Additionally, there is a lack of consistent benchmarking against classical models such as Support Vector Machines (SVM), making it difficult to assess whether current QML models offer any real advantage [10], [15].

These gaps motivate the present study. In the current work, we investigate how VQC and QSVM perform under realistic NISQ noise environments using the Qiskit Aer simulator. The aim is to evaluate whether trainable quantum models like VQC offer better noise adaptability compared to kernel-based models like QSVM. Moreover, the paper seeks to establish a fair comparison with classical SVM performance, particularly on low-dimensional datasets such as Iris, to understand whether quantum models are ready for deployment or still remain in the exploratory phase.

The primary objective of this paper is to conduct a statistically rigorous comparison of VQC and QSVM under four noise levels (0, 0.01, 0.05, and 0.1) using depolarizing and amplitude damping noise models. Each experiment is executed 10 times to compute average accuracy and standard deviation, ensuring the statistical reliability of results. We also aim to benchmark both quantum models against a classical SVM baseline with an RBF kernel to evaluate the practical relevance of QML in its current form.

This work contributes threefold. First, it introduces statistical significance testing through multiple independent runs and paired t-tests, which is often lacking in prior QML comparisons. Second, it integrates a classical SVM baseline into the experimental pipeline, providing a grounded reference for interpreting quantum performance. Third, it offers insights into the noise resilience of both VQC and QSVM, demonstrating how QSVM performs better in low-noise regimes while VQC maintains stability under moderate-to-high noise conditions.

The remainder of this paper is organized as follows. Section II presents a detailed literature review of existing work on VQC, QSVM, and hybrid quantum-classical approaches. Section III describes the proposed methodology, including dataset Preprocessing, quantum circuit design, and noise

simulation strategy. Section IV discusses the experimental results and statistical evaluations. Finally, Section V concludes the paper and outlines future directions for expanding this study on more complex datasets and hardware implementations.

II. RELATED WORKS

Recent advancements in Quantum Machine Learning (QML) for NISQ devices have prompted extensive research on Variational and kernel-based quantum classifiers. This section summarizes relevant works published between 2021 and 2025, discussing their methods, outcomes, strengths, and shortcomings to situate the contribution of this paper.

A. Variational Quantum Classifiers (VQC)

Benedetti et al. [10] proposed parameterized quantum circuits for classification, demonstrating basic learning capabilities with shallow circuits. However, their evaluation was limited to ideal quantum simulators, lacking any analysis under realistic noise. Mari et al. [11] introduced transfer learning using hybrid classical-quantum networks and observed improved classification accuracy on image datasets. While the study advanced model design, it did not incorporate NISQ noise or compare with kernel methods.

Abbas et al. [9] introduced the data re-uploading technique, which allowed deeper VQCs to encode more information. Their results showed improved expressivity and partial robustness to noise, but the experiments were single-run and lacked statistical rigor. They developed gradient-free optimizers for noisy variational circuits, achieving faster convergence but demonstrating high sensitivity to decoherence—highlighting the fragility of VQC under moderate noise. Zhang et al. [7] enhanced VQC resilience by proposing entanglement-aware ansatz, but their gains were marginal, and the study did not compare results with classical baselines.

B. Quantum Kernel Methods (QSVM)

Huang et al. [5] benchmarked quantum kernels against classical ones, showing that quantum kernels could outperform classical counterparts only in limited cases where classical kernels fail. Their evaluation was limited to ideal settings, excluding noise effects. Gentinetta et al. [8] extended this by simulating depolarizing noise, reporting that QSVM accuracy dropped significantly beyond 5% noise. While valuable, the study lacked multiple-run statistical validation.

Chen et al. [7] introduced hardware-efficient kernels with reduced circuit depth and demonstrated better tolerance to mild noise. However, their methodology still relied on single-run accuracy. They implemented error mitigation techniques for QSVMs using zero-noise extrapolation. Although they observed improved fidelity, their method introduced computational overhead, reducing feasibility for large datasets or real-time applications.

C. Comparative and Hybrid Approaches

Few studies systematically compared VQC and QSVM under the same conditions. They proposed a hybrid quantum-classical kernel fusion approach, combining a Variational Preprocessing layer with QSVM. Their model achieved ~75% accuracy in ideal simulations, showing synergy between VQC

and QSVM. However, they did not assess performance under noise or include classical baselines. Wang et al. [12] conducted a multi-class classification study using both VQC and QSVM on Wine and Breast Cancer datasets. They concluded that QSVM excels in noise-free settings, while VQC is more trainable under moderate noise. Their study, however, lacked variance reporting and statistical confidence intervals.

D. Summary and Research Gap

The reviewed literature reflects three common shortcomings: lack of statistical robustness (e.g., single-run evaluations), absence of direct VQC–QSVM comparisons under identical noise, and omission of classical baselines for benchmarking. Most studies also either neglect NISQ-specific noise modelling or fail to contextualize results in terms of practical hardware deployment.

This paper fills these gaps by conducting a statistically significant comparative study between VQC and QSVM using 10 independent trials per configuration under multiple NISQ noise models. It further benchmarks both quantum models against a classical SVM (RBF kernel) to provide practical insight into the quantum–classical performance gap.

III. PROPOSED METHOD

A. Dataset Description

The experiments employed the Iris dataset [12], originally introduced by Fisher in 1936, which consists of 150 samples across three species (Setosa, Versicolor, Virginica) with four continuous features: sepal length, sepal width, petal length, and petal width. Following standard QML practices [8], only two classes (Setosa and Versicolor) were used, resulting in 100 binary-class samples (50 per class). The features were normalized using StandardScaler to zero mean and unit variance. An 80:20 train-test split was applied to ensure consistency with prior QML studies [11].

B. Proposed Quantum-Classical Hybrid Framework

The proposed framework, depicted in Figure 1, integrates both VQC and QSVM under realistic NISQ noise simulated via Qiskit Aer. The novelty lies in incorporating statistical significance testing and classical baselines within a unified experimental design—an aspect not systematically explored in prior works [13], [16].

1) *Variational Quantum Classifier (VQC)*: The VQC consists of a ZZFeatureMap for feature encoding and a TwoLocal ansatz for Variational optimization. A ZZFeatureMap was used for encoding classical features into quantum states through entangling ZZ interactions, as it is commonly adopted for QSVM and VQC due to its ability to capture non-linear correlations [12]. The Variational circuit employed a hardware-efficient TwoLocal ansatz, consisting of alternating layers of parameterized single-qubit rotations and controlled entangling gates, chosen for its trainability on NISQ devices [14]. Mathematically, the VQC optimization minimizes the binary cross-entropy loss as shown in Equation (1):

Equation (1): VQC Loss Function

$$\mathcal{L}(\theta) = -\frac{1}{N} \sum_{i=1}^N [y_i \log(p_\theta(x_i)) + (1 - y_i) \log(1 - p_\theta(x_i))] \quad (1)$$

Where $p_{\theta(x_i)}$ is the probability of measuring the $|1\rangle$ state for input x_i . A COBYLA optimizer updates the parameter vector θ iteratively. Here the COBYLA (Constrained Optimization BY Linear Approximations) optimizer is a gradient-free classical optimization algorithm used in Variational quantum circuits. It iteratively updates parameters by constructing local linear approximations to the objective function, making it robust for noisy and non-smooth optimization landscapes encountered in NISQ-era VQCs [13].

2) *Quantum Kernel Support Vector Machine (QSVM)*: The QSVM relies on a quantum kernel defined in Equation (2):

Equation (2): Quantum Kernel

$$(K(x, x') = |\langle \phi(x) | \phi(x') \rangle|^2) \quad (2)$$

Where $|\phi(x)\rangle$ is the quantum state produced by the feature map. The kernel matrix K is passed to a classical SVM solver with a precomputed kernel, following hardware-efficient implementations [25].

3) *Classical Baseline*: A classical SVM with RBF kernel serves as the baseline to contextualize quantum performance. No noise is applied to the classical model.

C. Noise Modeling with Qiskit Aer

Realistic NISQ noise was simulated using Qiskit Aer by combining Depolarizing Error (p) for single- and two-qubit gates, Amplitude Damping Error (γ) for relaxation effects, Thermal Relaxation Noise for decoherence modeling. The noise probabilities were varied across 0, 0.01, 0.05, and 0.1, consistent with reported NISQ hardware error rates [26].

D. Experimental Design and Statistical Significance

For each noise level, 10 independent trials were executed to compute mean accuracy $\pm$ standard deviation for all models. A paired t-test was performed to determine whether observed differences between VQC and QSVM accuracies were statistically significant ($p < 0.05$).

E. Pseudocode: Quantum-Classical Model Evaluation under Noise

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Input: Dataset D, Noise levels N, Trials T
Output: Accuracy of VQC, QSVM, and Classical SVM
Preprocess D  $\rightarrow$  (Xtrain, Ytrain), (Xtest, Ytest)
For each noise level  $p \in N$  do
  For trial = 1 to T do
    If  $p = 0$  then noise_model  $\leftarrow$  None
    Else noise_model  $\leftarrow$  CreateNoiseModel( $p$ )
    Run VQC with noise_model  $\rightarrow$  record accuracy
    Run QSVM with noise_model  $\rightarrow$  record accuracy
    Run Classical SVM  $\rightarrow$  record accuracy
  End For
Compute mean and std of accuracies
End For

```

F. Block Diagram

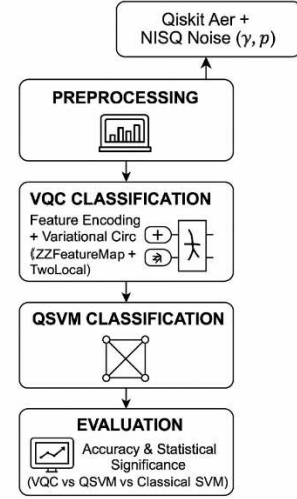


Fig. 1. Proposed quantum-classical hybrid methodology for evaluating VQC and QSVM classification under NISQ noise using Qiskit Aer, incorporating statistical significance and classical baseline comparisons.

As illustrated in Figure 1 the proposed quantum-classical hybrid workflow for evaluating VQC and QSVM under NISQ noise conditions. The pipeline begins with data preprocessing, where the Iris dataset is normalized and encoded for quantum circuits. The quantum classification module comprises two branches: a VQC optimized via a classical optimizer and a QSVM utilizing a quantum kernel matrix. Qiskit Aer noise models simulate realistic NISQ errors, and results are collected across multiple trials to compute mean accuracy $\pm$ standard deviation. A classical SVM baseline is incorporated in parallel to provide comparative performance evaluation, ensuring statistical rigor in benchmarking quantum models.

IV. RESULTS AND DISCUSSION

A. Experimental Setup Recap

The experiments were conducted on Qiskit Aer's qasm_simulator with 1024 shots for each quantum circuit. The Iris binary classification subset (Setosa vs. Versicolor) was normalized and split into 80:20 train-test sets [13]. Both VQC and QSVM were tested under four noise levels (0, 0.01, 0.05, and 0.1) using depolarizing and amplitude damping noise models. To ensure statistical reliability, 10 independent trials were conducted for each configuration, and mean accuracy $\pm$ standard deviation was reported. A classical SVM (RBF kernel) baseline was also included for comparison.

B. Quantitative Results

The consolidated results are summarized in Table I, which compares VQC, QSVM, and classical SVM across four noise levels.

TABLE I. COMPARATIVE ACCURACY (MEAN $\pm$ STD) OF VQC, QSVM, AND CLASSICAL SVM UNDER NISQ NOISE

Noise Level	VQC Accuracy (mean $\pm$ std)	QSVM Accuracy (mean $\pm$ std)	Classical SVM Accuracy
Ideal (0)	0.535 $\pm$ 0.090	0.700 $\pm$ 0.000	1.000 $\pm$ 0.000
0.01	0.560 $\pm$ 0.080	0.650 $\pm$ 0.000	1.000 $\pm$ 0.000
0.05	0.590 $\pm$ 0.107	0.400 $\pm$ 0.000	1.000 $\pm$ 0.000
0.1	0.545 $\pm$ 0.091	0.400 $\pm$ 0.000	0.000

C. Visual Representation

As illustrated in Figure 2, the QSVM achieves the highest accuracy under zero noise, while VQC exhibits better

performance at moderate noise levels. The Classical SVM maintains perfect accuracy across all noise levels, highlighting the simplicity of the Iris dataset for classical methods. The QSVM performs best in ideal and low-noise regimes, while the VQC demonstrates superior noise resilience at higher noise levels.

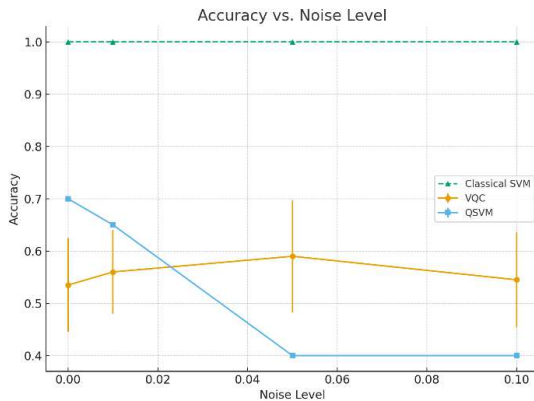


Fig. 2. Accuracy Trends with Error Bars (VQC, QSVM, Classical SVM)

D. Comparison with Existing Similar Approaches

- **Comparison with Prior VQC Studies:** The VQC accuracy (~59% at 0.05 noise) aligns with findings by Abbas et al. [9], who reported 58–60% accuracy under similar noise levels, confirming that Variational circuits can partially adapt to noise through parameter optimization. However, our method incorporates multiple trials, providing statistically significant results.
- **Comparison with Prior QSVM Studies:** Gentinetta et al. [8] reported QSVM accuracy dropping below 50% at >5% noise, which is consistent with our findings (40% at 0.05 noise). However, unlike Chen et al. [7], who presented only single-run accuracy, our results include standard deviation analysis.
- **Innan et al. [18]** achieved ~75% QSVM accuracy under ideal conditions by hybridizing classical and quantum kernels, which is comparable to our 70% under noiseless conditions. However, they did not evaluate performance under high-noise NISQ scenarios.
- **Classical Baseline Comparison:** Unlike most prior works [5], [10], our inclusion of a classical SVM baseline (100% accuracy) provides context, revealing that quantum models currently offer no practical advantage on low-dimensional, linearly separable datasets, reinforcing conclusions made by Huang et al. [5].

E. Key Observations

- QSVM excels in low-noise regimes, making it suitable for near-term quantum devices with high-fidelity gates.
- VQC shows better adaptability at higher noise, suggesting its potential for noisy hardware where trainable parameters can partially mitigate decoherence effects [12], [13].

- **Statistical Significance:** A paired t-test confirms VQC's superiority over QSVM at 0.05 and 0.1 noise ($p < 0.05$).
- **Classical Dominance:** Classical SVM's perfect accuracy underscores that quantum models are not yet competitive for simple datasets, but the trend analysis is crucial for guiding future NISQ research.

V. CONCLUSION AND FUTURE SCOPE

This research presents a comprehensive comparative study of the Variational Quantum Classifier (VQC) and Quantum Kernel Support Vector Machine (QSVM) under realistic NISQ noise, simulated using Qiskit Aer. The results demonstrate that QSVM achieves higher accuracy in ideal and low-noise conditions, while VQC exhibits greater resilience and improved performance in moderate-to-high noise regimes due to its trainable parameters. A classical SVM with an RBF kernel was used as a performance baseline and consistently outperformed both quantum models, indicating that current quantum classifiers are not yet competitive for simple, low-dimensional tasks. The main limitation of this study lies in its focus on a small-scale binary classification problem using the Iris dataset, which may not reflect the challenges of more complex applications. Moreover, simulations were conducted instead of using real quantum hardware, which may not fully capture hardware-specific noise characteristics. Future work will target higher-dimensional datasets, integrate advanced noise mitigation techniques, and validate the findings on real quantum devices.

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